PalmBridge Retail Co. — E-Commerce Sales and Profitability Analysis Project

1. Project Title

PalmBridge Retail Co. — E-Commerce Sales, Profitability and Customer Analysis

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2. Project Overview

This project focuses on analyzing the sales, profit, customers, products, and regional performance of **PalmBridge Retail Co.**, a fictional US-based online retail company.

The dataset contains three years of transactional e-commerce data from **January 2022 to December 2024**.

The dataset is intentionally unclean and includes common real-world data problems such as:

- Duplicate records
- Missing values
- Incorrect data types
- Inconsistent text values
- Invalid dates
- Incorrect numerical values
- Potentially negative or near-zero profit
- Formatting issues

The company generates more than **$1.6 million in sales**, but its overall profit is close to zero. The main objective of this project is to identify the reasons behind the low profitability and provide useful business recommendations.

> **Note:** This is a synthetic dataset created for learning and portfolio purposes. It does not represent a real company, customer, or transaction.

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3. Project Type

End-to-End Data Analysis Project

The project includes:

- Data Cleaning

- Exploratory Data Analysis
- Data Validation
- Feature Engineering
- Business Analysis
- SQL Analysis
- Power BI Dashboard Development
- Business Insights
- Recommendations

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4. Business Domain

E-Commerce and Retail Analytics

The project covers:

- Sales performance
- Profitability
- Product performance
- Customer segmentation
- Regional analysis
- Discount analysis
- Order analysis
- Time-based sales trends

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5. Business Problem

PalmBridge Retail Co. operates across five regions in the United States:

- North
- South
- East
- West
- Central

The company sells products across three major categories:

1. Technology
2. Furniture
3. Office Supplies

Although the company generates more than **$1.6 million in sales**, it records almost no profit.

The management wants to understand:

- Which products generate high sales but low profit?
- Which categories are causing losses?
- Are discounts affecting profitability?
- Which regions perform better?
- Which customer segments are more profitable?
- Which sub-categories should be promoted or discontinued?
- How has business performance changed between 2022 and 2024?

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6. Project Objectives

The main objectives of this project are:

1. Clean and prepare the raw e-commerce dataset.
2. Identify and remove duplicate records.
3. Handle missing and invalid values.
4. Analyze total sales and total profit.
5. Identify profitable and loss-making products.
6. Compare sales performance across regions.
7. Analyze customer segments.
8. Study the effect of discounts on profit.
9. Identify the best and worst-performing categories.
10. Analyze monthly and yearly sales trends.
11. Create SQL queries for business reporting.
12. Build an interactive Power BI dashboard.
13. Provide business recommendations to improve profitability.

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7. Dataset Description

7.1 Dataset Name

PalmBridge Retail Co. Synthetic E-Commerce Dataset

7.2 Dataset Type

Uncleaned transactional sales dataset.

7.3 Dataset Duration

January 2022 to December 2024

7.4 Number of Rows

The dataset contains:

- **501 rows**
- **1 intentional duplicate**
- Approximately **500 original transactional records**

7.5 Business Location

United States

7.6 Regions

The dataset includes five regions:

- North
- South
- East
- West
- Central

7.7 Customer Segments

The dataset contains three customer segments:

- Consumer
- Corporate
- Home Office

7.8 Product Categories

The dataset includes three product categories:

- Technology
- Furniture
- Office Supplies

7.9 Product Sub-Categories

The dataset contains fifteen product sub-categories.

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8. Expected Dataset Columns

The exact column names should be checked after loading the file. A typical retail dataset may contain columns similar to the following:

Column	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Shipping method
Customer ID	Unique customer identifier
Customer Name	Customer name
Segment	Customer segment
Country	Country of the customer
City	Customer city
State	Customer state
Region	Business region
Product ID	Unique product identifier
Category	Main product category
Sub-Category	Product sub-category
Product Name	Product name
Sales	Revenue generated from the order
Quantity	Number of products sold
Discount	Discount percentage
Profit	Profit or loss generated from the order

> The actual column names must be verified using `df.columns` because the dataset may contain intentionally modified or inconsistent column names.

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9. Technology Stack

Programming and Analysis

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Database

- MySQL or SQLite
- SQL

Dashboard

- Power BI
- DAX

Documentation

- Excel
- PowerPoint or PDF
- GitHub README

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10. Recommended Project Folder Structure

```
PalmBridge_Retail_Analysis/
│
├── data/
│   ├── raw/
│   │   └── palmbridge_retail_uncleaned.csv
│   │
│   └── processed/
```

```
│       └── palmbridge_retail_cleaned.csv
│
├── notebooks/
│   ├── 01_data_understanding.ipynb
│   ├── 02_data_cleaning.ipynb
│   ├── 03_exploratory_analysis.ipynb
│   └── 04_business_insights.ipynb
│
├── sql/
│   └── palmbridge_analysis.sql
│
├── powerbi/
│   └── palmbridge_dashboard.pbix
│
├── reports/
│   └── palmbridge_project_report.pdf
│
├── images/
│   └── dashboard_screenshots/
│
├── README.md
└── requirements.txt
```

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11. End-to-End Project Workflow

Business Problem
↓
Load Raw Dataset
↓
Understand Dataset
↓
Check Data Quality
↓
Clean Dataset
↓
Validate Data
↓
Feature Engineering
↓
Exploratory Data Analysis
↓
Business Analysis
↓
SQL Analysis
↓

Power BI Dashboard
↓
Insights and Recommendations
↓
Documentation
↓
GitHub Portfolio

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12. Step 1 — Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

---

13. Step 2 — Load the Dataset

```
df = pd.read_csv("palmbridge_retail_uncleaned.csv")
```

Display the first five rows:

```
df.head()
```

Display the last five rows:

```
df.tail()
```

---

14. Step 3 — Understand the Dataset

Check the number of rows and columns

```
df.shape
```

Expected result:

(501, number_of_columns)

Check column names

```
df.columns
```

Check data types

df.dtypes

Get dataset information

df.info()

Generate statistical summary

df.describe(include="all")

Check unique values

df.nunique()

---

15. Step 4 — Check Missing Values

df.isnull().sum()

Percentage of missing values:

```
missing_percentage = (
    df.isnull().sum() / len(df)
) * 100

missing_percentage.sort_values(ascending=False)
```

Visualize missing values:

```
plt.figure(figsize=(12, 6))
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values in PalmBridge Dataset")
plt.show()
```

---

16. Step 5 — Check Duplicate Records

Count duplicate rows

df.duplicated().sum()

Expected result:

1

Display duplicate records

df[df.duplicated()]

Remove duplicate records

df = df.drop_duplicates()

Verify duplicates again

df.duplicated().sum()

Expected result:

0

---

17. Step 6 — Clean Column Names

Convert column names to lowercase:

```
df.columns = df.columns.str.lower()
```

Replace spaces with underscores:

```
df.columns = (
    df.columns
    .str.strip()
    .str.replace(" ", "_")
)
```

Example:

Order Date → order_date
Customer Name → customer_name
Sub-Category → sub-category

To replace hyphens:

```
df.columns = df.columns.str.replace("-", "_")
```

---

18. Step 7 — Clean Text Columns

Identify text columns:

text_columns = df.select_dtypes(include="object").columns

Remove extra spaces:

```
for column in text_columns:
    df[column] = df[column].str.strip()
```

Convert selected columns to a consistent format:

```
df["region"] = df["region"].str.title()
df["category"] = df["category"].str.title()
df["segment"] = df["segment"].str.title()
```

Check unique values:

```
df["region"].unique()
df["category"].unique()
df["segment"].unique()
```

---

19. Step 8 — Convert Date Columns

Convert order date into datetime format:

```
df["order_date"] = pd.to_datetime(
    df["order_date"],
    errors="coerce"
)
```

If a shipping date exists:

```
df["ship_date"] = pd.to_datetime(
    df["ship_date"],
    errors="coerce"
)
```

Check invalid dates:

```
df["order_date"].isnull().sum()
```

---

20. Step 9 — Convert Numerical Columns

```
numeric_columns = [
    "sales",
```

```
    "quantity",
    "discount",
    "profit"
]
for column in numeric_columns:
    df[column] = pd.to_numeric(
        df[column],
        errors="coerce"
    )
```

Check numerical data types:

```
df[numeric_columns].dtypes
```

---

21. Step 10 — Validate the Dataset

Check invalid sales values

```
df[df["sales"] < 0]
```

Check invalid quantity values

```
df[df["quantity"] <= 0]
```

Check discount range

```
df["discount"].describe()
```

If discount is stored as a percentage:

```
df["discount"].unique()
```

Check profit values

```
df["profit"].describe()
```

Check date range

```
df["order_date"].min(), df["order_date"].max()
```

Expected date range:

2022-01-01 to 2024-12-31

---

22. Step 11 — Handle Missing Values

First, identify missing values:

df.isnull().sum()

Possible approaches:

Numerical columns

df["sales"] = df["sales"].fillna(df["sales"].median())

Categorical columns

df["region"] = df["region"].fillna("Unknown")

Date columns

df = df.dropna(subset=["order_date"])

> The correct treatment depends on the business meaning of each missing value.
> Missing values should not be filled blindly.

---

23. Step 12 — Feature Engineering

Create additional columns to improve analysis.

Extract year

df["year"] = df["order_date"].dt.year

Extract month

df["month"] = df["order_date"].dt.month

Extract month name

df["month_name"] = df["order_date"].dt.month_name()

Extract quarter

df["quarter"] = df["order_date"].dt.quarter

Extract weekday

```
df["weekday"] = df["order_date"].dt.day_name()
```

Calculate shipping duration

If `ship_date` is available:

```
df["shipping_days"] = (
    df["ship_date"] - df["order_date"]
).dt.days
```

Create profit status

```
df["profit_status"] = np.where(
    df["profit"] > 0,
    "Profitable",
    np.where(
        df["profit"] < 0,
        "Loss",
        "Break-even"
    )
)
```

Create discount category

```
df["discount_category"] = pd.cut(
    df["discount"],
    bins=[-0.01, 0, 0.10, 0.30, 1],
    labels=[
        "No Discount",
        "Low Discount",
        "Medium Discount",
        "High Discount"
    ]
)
```

---

24. Step 13 — Basic Business Metrics

Total sales

```
total_sales = df["sales"].sum()
total_sales
```

Total profit

```
total_profit = df["profit"].sum()
total_profit
```

Total quantity sold

```
total_quantity = df["quantity"].sum()
total_quantity
```

Total orders

```
total_orders = df["order_id"].nunique()
total_orders
```

Total customers

```
total_customers = df["customer_id"].nunique()
total_customers
```

Average order value

```
average_order_value = total_sales / total_orders
average_order_value
```

Profit margin

```
profit_margin = (
    total_profit / total_sales
) * 100

profit_margin
```

---

25. Step 14 — Sales Analysis by Category

```
category_sales = (
    df.groupby("category")["sales"]
    .sum()
    .sort_values(ascending=False)
)

category_sales
```

Visualize category sales:

```
category_sales.plot(
    kind="bar",
    figsize=(8, 5),
    title="Sales by Product Category"
)

plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.show()
```

---

26. Step 15 — Profit Analysis by Category

```
category_profit = (
    df.groupby("category")["profit"]
    .sum()
    .sort_values(ascending=False)
)

category_profit
```

Compare sales and profit:

```
category_summary = df.groupby("category").agg(
    total_sales=("sales", "sum"),
    total_profit=("profit", "sum"),
    total_quantity=("quantity", "sum"),
    average_discount=("discount", "mean")
)

category_summary
```

This analysis helps identify categories that generate:

- High sales and high profit
- High sales and low profit
- Low sales and high profit margin
- High sales and negative profit

---

27. Step 16 — Regional Analysis

Sales by region

```
region_sales = (
    df.groupby("region")["sales"]
    .sum()
    .sort_values(ascending=False)
)

region_sales
```

Profit by region

```
region_profit = (
    df.groupby("region")["profit"]
    .sum()
    .sort_values(ascending=False)
)

region_profit
```

Regional summary

```
region_summary = df.groupby("region").agg(
    total_sales=("sales", "sum"),
    total_profit=("profit", "sum"),
    total_orders=("order_id", "nunique"),
    average_discount=("discount", "mean")
)

region_summary
```

---

28. Step 17 — Customer Segment Analysis

Sales by customer segment

```
segment_sales = (
    df.groupby("segment")["sales"]
    .sum()
    .sort_values(ascending=False)
)

segment_sales
```

Profit by customer segment

```
segment_profit = (
    df.groupby("segment")["profit"]
    .sum()
    .sort_values(ascending=False)
)

segment_profit
```

Customer segment summary

```
segment_summary = df.groupby("segment").agg(
    total_sales=("sales", "sum"),
    total_profit=("profit", "sum"),
    total_orders=("order_id", "nunique"),
    average_discount=("discount", "mean")
)

segment_summary
```

---

29. Step 18 — Sub-Category Analysis

```
subcategory_summary = df.groupby("sub_category").agg(
    total_sales=("sales", "sum"),
    total_profit=("profit", "sum"),
    total_quantity=("quantity", "sum"),
    average_discount=("discount", "mean")
).sort_values(
    by="total_profit",
    ascending=False
)

subcategory_summary
```

Most profitable sub-categories

```
subcategory_summary.head(10)
```

Loss-making sub-categories

```
subcategory_summary[
    subcategory_summary["total_profit"] < 0
]
```

---

30. Step 19 — Discount and Profitability Analysis

Discounts may increase sales but reduce profit.

Average profit by discount

```
discount_profit = df.groupby("discount").agg(
    total_sales=("sales", "sum"),
    total_profit=("profit", "sum"),
    average_profit=("profit", "mean")
)

discount_profit
```

Correlation between discount and profit

```
df[["discount", "profit"]].corr()
```

Visualize discount and profit

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="discount",
    y="profit"
)

plt.title("Relationship Between Discount and Profit")
plt.show()
```

Business questions:

- Does a higher discount result in lower profit?
- Which categories receive the highest discounts?
- Are loss-making products heavily discounted?
- Which customer segments receive more discounts?

---

31. Step 20 — Time-Based Analysis

Yearly sales

```
yearly_sales = df.groupby("year")["sales"].sum()
yearly_sales
```

Yearly profit

```
yearly_profit = df.groupby("year")["profit"].sum()
yearly_profit
```

Monthly sales

```
monthly_sales = (
    df.groupby(["year", "month"])["sales"]
    .sum()
    .reset_index()
)

monthly_sales
```

Sales trend

```
plt.figure(figsize=(10, 5))

sns.lineplot(
    data=monthly_sales,
    x="month",
    y="sales",
    hue="year",
    marker="o"
)

plt.title("Monthly Sales Trend")
plt.show()
```

---

32. Step 21 — Top Products Analysis

Top products by sales

```
top_products_sales = (
    df.groupby("product_name")["sales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
```

```
top_products_sales
```

Top products by profit

```
top_products_profit = (
    df.groupby("product_name")["profit"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_products_profit
```

Loss-making products

```
product_profit = (
    df.groupby("product_name")["profit"]
    .sum()
    .sort_values()
)

product_profit.head(10)
```

---

33. Step 22 — Customer-Level Analysis

Top customers by sales

```
top_customers = (
    df.groupby("customer_name")["sales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_customers
```

Customer profitability

```
customer_profit = (
    df.groupby("customer_name")["profit"]
    .sum()
    .sort_values(ascending=False)
)
```

customer_profit

Customer order frequency

```
customer_orders = (
    df.groupby("customer_id")["order_id"]
    .nunique()
    .sort_values(ascending=False)
)

customer_orders.head(10)
```

---

34. Step 23 — Profitability Classification

Create a classification based on sales and profit.

```
df["business_status"] = np.select(
    [
        df["profit"] > 0,
        df["profit"] < 0,
        df["profit"] == 0
    ],
    [
        "Profitable",
        "Loss-making",
        "Break-even"
    ],
    default="Unknown"
)
```

Count each status:

```
df["business_status"].value_counts()
```

Visualize:

```
sns.countplot(
    data=df,
    x="business_status"
)

plt.title("Profitability Status of Transactions")
plt.show()
```

---

35. Step 24 — SQL Database Design

The cleaned dataset can be loaded into a SQL table named:

palmbridge_sales

Suggested table columns:

row_id
order_id
order_date
ship_date
ship_mode
customer_id
customer_name
segment
country
city
state
region
product_id
category
sub_category
product_name
sales
quantity
discount
profit
year
month
quarter
profit_status

---

36. Step 25 — SQL Business Questions

Total sales

```
SELECT
    SUM(sales) AS total_sales
FROM palmbridge_sales;
```

Total profit

```
SELECT
```

```
    SUM(profit) AS total_profit
FROM palmbridge_sales;
```

Sales by category

```
SELECT
    category,
    SUM(sales) AS total_sales
FROM palmbridge_sales
GROUP BY category
ORDER BY total_sales DESC;
```

Profit by category

```
SELECT
    category,
    SUM(profit) AS total_profit
FROM palmbridge_sales
GROUP BY category
ORDER BY total_profit DESC;
```

Sales by region

```
SELECT
    region,
    SUM(sales) AS total_sales
FROM palmbridge_sales
GROUP BY region
ORDER BY total_sales DESC;
```

Loss-making products

```
SELECT
    product_name,
    SUM(profit) AS total_profit
FROM palmbridge_sales
GROUP BY product_name
HAVING SUM(profit) < 0
ORDER BY total_profit;
```

Top ten products by sales

```
SELECT
    product_name,
    SUM(sales) AS total_sales
FROM palmbridge_sales
GROUP BY product_name
```

```
ORDER BY total_sales DESC
LIMIT 10;
```

Average discount by category

```
SELECT
    category,
    AVG(discount) AS average_discount
FROM palmbridge_sales
GROUP BY category;
```

Yearly sales

```
SELECT
    YEAR(order_date) AS order_year,
    SUM(sales) AS total_sales
FROM palmbridge_sales
GROUP BY YEAR(order_date)
ORDER BY order_year;
```

Customer segment performance

```
SELECT
    segment,
    SUM(sales) AS total_sales,
    SUM(profit) AS total_profit,
    COUNT(DISTINCT order_id) AS total_orders
FROM palmbridge_sales
GROUP BY segment;
```

High-discount transactions

```
SELECT *
FROM palmbridge_sales
WHERE discount >= 0.30;
```

---

37. Step 26 — Power BI Data Preparation

Before importing the data into Power BI:

1. Remove duplicates.
2. Correct data types.
3. Convert date columns to Date type.
4. Check missing values.
5. Rename columns.

6. Create calculated columns.
7. Validate sales and profit values.
8. Create a proper date table.

---

38. Step 27 — Power BI Data Model

The main table can be named:

FactSales

Recommended supporting tables:

DimDate
DimCustomer
DimProduct
DimRegion
DimCategory

Possible model:

```
DimDate
  |
FactSales
  |
DimProduct
  |
DimCustomer
  |
DimRegion
```

---

39. Step 28 — Important DAX Measures

Total Sales

```
Total Sales =
SUM(FactSales[sales])
```

Total Profit

```
Total Profit =
SUM(FactSales[profit])
```

Total Quantity

```
Total Quantity =
SUM(FactSales[quantity])
```

Total Orders

```
Total Orders =
DISTINCTCOUNT(FactSales[order_id])
```

Total Customers

```
Total Customers =
DISTINCTCOUNT(FactSales[customer_id])
```

Average Order Value

```
Average Order Value =
DIVIDE(
    [Total Sales],
    [Total Orders]
)
```

Profit Margin

```
Profit Margin =
DIVIDE(
    [Total Profit],
    [Total Sales]
)
```

Average Discount

```
Average Discount =
AVERAGE(FactSales[discount])
```

Loss-Making Orders

```
Loss-Making Orders =
CALCULATE(
    [Total Orders],
    FactSales[profit] < 0
)
```

---

40. Step 29 — Power BI Dashboard Pages

Page 1: Executive Overview

Include:

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Total Customers
- Total Quantity
- Sales by Year
- Profit by Year
- Sales by Region

Page 2: Category and Product Analysis

Include:

- Sales by Category
- Profit by Category
- Sales by Sub-Category
- Top 10 Products by Sales
- Top 10 Products by Profit
- Loss-Making Products

Page 3: Regional Performance

Include:

- Sales by Region
- Profit by Region
- Orders by Region
- Regional Profit Margin
- Region-wise Category Performance

Page 4: Customer Analysis

Include:

- Sales by Customer Segment
- Profit by Customer Segment

- Number of Customers
- Average Order Value
- Customer Sales Ranking

Page 5: Discount and Profit Analysis

Include:

- Discount Distribution
- Average Discount by Category
- Discount vs Profit Scatter Plot
- Profit by Discount Category
- High-Discount Loss-Making Products

Page 6: Time-Based Analysis

Include:

- Monthly Sales Trend
- Monthly Profit Trend
- Yearly Sales Comparison
- Quarterly Sales
- Weekday Performance

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41. Step 30 — Important Dashboard KPIs

The dashboard should display the following KPIs:

KPI	Purpose
Total Sales	Measures total revenue
Total Profit	Measures business profitability
Profit Margin	Measures profit efficiency
Total Orders	Measures order volume
Total Customers	Measures customer reach
Total Quantity	Measures product volume
Average Order Value	Measures average order revenue

Average Discount	Measures discount usage
Loss-Making Orders	Measures unprofitable transactions
Top Category	Identifies the strongest category
Top Region	Identifies the strongest region
Top Product	Identifies the best-selling product

---

42. Step 31 — Key Business Questions

The final project should answer the following questions:

Sales Performance

1. What is the total sales value?
2. Which year generated the highest sales?
3. Which month generated the highest sales?
4. Which category contributes the most revenue?
5. Which region generates the highest sales?

Profitability

6. What is the total profit?
7. Why is profit close to zero?
8. Which categories are loss-making?
9. Which products generate high sales but low profit?
10. Which regions have the lowest profit margin?

Discount Analysis

11. Does discounting reduce profit?
12. Which categories receive the highest discounts?
13. Are heavily discounted products loss-making?
14. Which customer segment receives the highest discount?

Customer Analysis

15. Which customer segment generates the most sales?
16. Which segment generates the highest profit?
17. Who are the top customers?
18. What is the average order value by segment?

Product Analysis

19. Which products are best-selling?
20. Which products are most profitable?
21. Which products generate losses?
22. Which sub-categories require business attention?

---

43. Step 32 — Possible Business Insights

The following are examples of insights that may be identified after completing the analysis. They should be confirmed using actual calculations.

- A category may generate high sales but contribute very little profit.
- Some products may have high discounts and negative profit.
- One region may generate strong revenue but weak profitability.
- Corporate customers may generate higher order values.
- Office Supplies may have lower sales but better margins.
- Furniture products may require additional shipping or discount analysis.
- A small number of products may contribute a large percentage of total sales.
- Loss-making products may need pricing or discount adjustments.
- Sales may increase over time while profit remains stagnant.
- Some customer segments may be valuable in terms of sales but less profitable.

---

44. Step 33 — Business Recommendations

Based on the final analysis, PalmBridge Retail Co. may consider the following actions:

1. Review High-Discount Products

Reduce unnecessary discounts on products that already generate low or negative profit.

2. Improve Product Pricing

Review pricing for products with high sales but weak profit margins.

3. Focus on Profitable Categories

Promote categories and sub-categories that generate strong profit margins.

4. Review Loss-Making Products

Investigate products with repeated negative profit.

5. Improve Regional Strategy

Study regions with high sales but low profitability.

6. Segment Customer Offers

Provide targeted discounts based on customer profitability rather than offering the same discount to everyone.

7. Monitor Profit Margin Regularly

Track profit margin along with sales instead of focusing only on revenue.

8. Reduce Unnecessary Discounts

Create discount limits for products with already-low margins.

9. Improve Product Mix

Promote products that provide a better balance between sales volume and profit.

10. Build a Regular Reporting System

Use Power BI to monitor sales, profit, discounts, and regional performance every month.

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45. Step 34 — Data Limitations

The project has the following limitations:

1. The dataset is synthetic.
2. It does not represent a real company.
3. It does not contain real customer information.

4. The dataset contains only 501 rows.
5. The dataset may not represent the complete retail business.
6. The results should not be treated as actual business performance.
7. Some missing values may be intentionally introduced.
8. The dataset contains one intentional duplicate.
9. The business context is fictional.
10. Recommendations are based only on the available columns.

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46. Step 35 — Final Project Deliverables

The completed project should include:

- Cleaned CSV file
- Python notebook
- Data cleaning report
- Exploratory data analysis
- SQL queries
- Power BI dashboard
- Business insights
- Business recommendations
- Project report
- GitHub repository
- README file
- Dashboard screenshots

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47. Suggested GitHub README Structure

PalmBridge Retail Co. E-Commerce Sales Analysis

Project Overview

Business Problem

Objectives

Dataset Description

Tools and Technologies

Data Cleaning

Exploratory Data Analysis

SQL Analysis

Power BI Dashboard

Key Insights

Business Recommendations

Project Structure

How to Run the Project

Limitations

Conclusion

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48. Resume Project Description

PalmBridge Retail Co. — E-Commerce Sales and Profitability Analysis

- Cleaned and analyzed a synthetic e-commerce dataset containing 500+ transactional records from 2022–2024 using Python and Pandas.
- Performed data cleaning, duplicate removal, missing-value handling, feature engineering, and exploratory data analysis.
- Analyzed sales, profit, discounts, customer segments, product categories, sub-categories, and regional performance.
- Used SQL to generate business insights and developed an interactive Power BI dashboard with KPIs for sales, profit, orders, customers, and profit margin.

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49. Interview Explanation

> “I worked on an e-commerce sales and profitability analysis project for a fictional company called PalmBridge Retail Co. The dataset contained three years of unclean transactional data with 501 rows, including one intentional duplicate. My main objective was to understand why the company generated more than $1.6 million in sales but almost no profit. I used Python and Pandas for data cleaning, handled duplicates and missing values, converted date and numerical columns, and created features such as year, month, quarter, profit status, and discount

category. I then analyzed sales and profit by category, region, customer segment, and product. I also used SQL for business queries and Power BI to build an interactive dashboard. The main focus of the project was identifying loss-making products, understanding the effect of discounts, and providing recommendations to improve profitability."

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50. Final Project Checklist

Data Understanding

- Loaded the dataset
- Checked rows and columns
- Checked data types
- Checked unique values
- Checked missing values
- Checked duplicate records

Data Cleaning

- Removed duplicate rows
- Cleaned column names
- Removed extra spaces
- Corrected text values
- Converted date columns
- Converted numerical columns
- Handled missing values
- Validated numerical values

Data Analysis

- Calculated total sales
- Calculated total profit
- Calculated profit margin
- Analyzed categories
- Analyzed sub-categories
- Analyzed regions
- Analyzed customer segments
- Analyzed discounts
- Analyzed yearly trends
- Analyzed monthly trends
- Identified top products
- Identified loss-making products

SQL

- Created SQL table
- Loaded cleaned data
- Wrote sales queries
- Wrote profit queries
- Wrote category queries
- Wrote regional queries
- Wrote customer queries
- Wrote product queries

Power BI

- Imported cleaned data
- Created data model
- Created DAX measures
- Created KPI cards
- Created charts
- Added slicers
- Created dashboard pages
- Validated dashboard values

Documentation

- Created project report
- Added dashboard screenshots
- Created GitHub README
- Added business insights
- Added recommendations
- Added resume description
- Prepared interview explanation

Conclusion

The **PalmBridge Retail Co. E-Commerce Sales and Profitability Analysis** project provides an end-to-end opportunity to practice real-world data analyst activities using a deliberately messy dataset.

The project focuses not only on calculating sales but also on understanding the most important business problem:

> **Why does the company generate high sales but almost no profit?**

By using Python, SQL, and Power BI, the project can identify loss-making products, inefficient discounting, weak regional performance, and customer segments that require further attention.